AYANA K K

RESEARCH ASSOCIATE | AGENTIC AI & EDGE AI

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in **Large Language Models, Agentic AI,** and **Edge AI**. Proven ability to design, optimize, and deploy **intelligent systems** that automate complex workflows, enable accurate knowledge retrieval at scale, and perform efficiently on resource-constrained hardware. Effective communicator, able to **translate complex technical concepts** for both technical teams and business stakeholders. Committed to building efficient, scalable, production-ready AI solutions, that drives measurable business value.

CORE SKILLS

- **Programming Languages:** Python, SQL, C/C++
- **AI & LLMs:** Agentic AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, AI Workflows
- **Retrieval & Vector Search:** FAISS (Flat, IVF, HNSW), Cross-Encoder Re-ranking, Retrieval Evaluation
- **Deep Learning & Edge AI:** PyTorch, Keras, CNNs, Transfer Learning, Model Compression, Pruned CNNs, BNNs, AdderNet, DeepShift
- **Machine Learning & Computer Vision:** Scikit-learn, XGBoost, Ensemble Learning, Feature Engineering, Semantic Segmentation, Object Detection
- **Backend & Media Processing:** FastAPI, REST APIs, Async Python (asyncio), React, ffmpeg, ElevenLabs
- **DevOps & Testing:** Docker, Docker Compose, GitHub Actions CI, pytest, Structured Logging, Caching
- **Data & Databases:** Pandas, NumPy, Matplotlib, Power BI, MySQL

EXPERIENCE

- **Research Associate**

  ReyraAI, Kozhikode | Feb 2026 – Present
  - Architected a **full-stack asynchronous AI video dubbing platform (FastAPI, React)** with a job-based REST API and async status polling, supporting multi-minute GenAI workloads without client timeouts. Implemented **concurrent multi-language processing** and fault-isolated pipelines, reducing batch turnaround by [X]%, and automated **frame-accurate ffmpeg editing, AI voice-track muxing, and SRT/VTT subtitle generation**.
  - Designed and deployed a **production-grade Retrieval-Augmented Generation (RAG) system** using LangChain, FAISS, and Ollama-hosted LLMs, containerized with Docker Compose and served via FastAPI. Optimized vector search (IVF/HNSW indexing) with **cross-encoder re-ranking** to cut query latency by [X]% and lift Recall@k by [X]%, backed by structured logging, caching, and CI/CD pipelines (pytest, GitHub Actions).
  - Developing **LLM-powered Agentic AI systems** with multi-step reasoning, tool calling, and autonomous workflow orchestration, automating complex decision-making processes and reducing manual intervention by [X]%.
  - Engineered energy-efficient **deep learning architectures for Edge AI,** including Pruned CNNs, Binary Neural Networks (BNNs), AdderNet, and DeepShift, applying model compression (structured pruning, binarization, shift-based quantization) to shrink model footprint by [X]% and accelerate inference [X]x with minimal accuracy loss.
  - Built a **reproducible PyTorch benchmarking framework** evaluating accuracy, latency, FLOPs, memory, and energy efficiency across [X] model variants, driving hardware-aware architecture selection for low-power, on-device AI deployment.

- **Data Annotation Specialist**

  Iva Technos, Coimbatore | Oct 2025 – Dec 2025
  - Onboarded rapidly to **advanced annotation workflows** using GT Studio, contributing to **Autonomous Driving Computer Vision** datasets.
  - Executed 3D bounding boxes, semantic segmentation, lane marking, and multi-class labeling ensuring consistent ground truth quality.

- **Data Science & AI Trainer**

  Maitexa Technology, Kozhikode | Jun 2025 – Aug 2025
  - Conducted hands-on training sessions on Python, Data Analysis, Machine Learning, and Deep Learning concepts.
  - Mentored students on data preprocessing, feature engineering, visualization, and model evaluation workflows.

EDUCATION

Master of Science — Computer Science (Intelligent Systems)
Central University of Kerala | Jul 2023 – Jun 2025

Bachelor of Science — Computer Science
Mahatma Gandhi College, Iritty (Kannur University) | May 2019 – May 2022
